

lost class:

direct proof, $P \Rightarrow Q$

some times, $P \Rightarrow Q$

Proof: Assume, P holds

- Definitions
- algebra
- logic

that is exactly what Q is:

Proposition:

a, b and c integers, if $a|b$

recall lost class defn

$b = k_1 a$

$k_1 = \text{integer}$

if a, b and c integers, if $a|b$ and $b|c$ then, $a|c$.

✓ $P: (a, b, c \in \mathbb{I}) \wedge (a|b) \wedge (b|c)$

✓ Q: $(a|c)$

Proof: suppose $(P \text{ true})$ $a, b, c \in \mathbb{Z}$, and $a|b$
and $b|c$

from Defⁿ, $a|b \Rightarrow b = k_1 a$ $[k_1: \text{integer}]$

also, Defⁿ, $b|c \Rightarrow c = k_2 b$ $[k_2: \text{integer}]$

substitution, $c = k_2 \cdot (k_1 a)$

$$c = (k_2 \cdot k_1) a$$

since, k_1 and k_2 are integers, so,

$k_1 \cdot k_2$ is also integer,

$\therefore k := k_1 \cdot k_2$ is also an integer

$\therefore c = k a$ $[k \text{ is integer}]$

$\therefore a|c$ $[from \text{ def}^n]$

• Division Algorithm:

For all integer a and $m > 0$

there exist unique q and r such that,

$$a = m \cdot q + r \quad 0 \leq r < m$$

Cases: $a = -105$ $m = 4$

there is a unique q and r

$$\boxed{q = -2, r = 3} \rightarrow \text{only choice}$$

Scenario ~~=~~ Cases

Proof using cases:

sometimes propositions require examining multiple cases,

Proposition: if n is integer, $n^2 + n + 6$ is even

P : n is integer, Q : $n^2 + n + 6$ is even,

Cases:

$n = \text{even}$, $n = \text{odd}$,

Case 1: suppose, n is even

from Defn, $n = 2k$ [k : integer]

Now,

$$\boxed{n^2 + n + 6} = (2k)^2 + (2k) + 6$$

$$= \boxed{4k^2 + 2k + 6} \rightarrow (\text{integer})$$

$$= 2 \boxed{(2k^2 + k + 3)} \rightarrow \text{integer}$$

we take it

$\therefore$

$$\therefore P = Q$$

$\therefore P$ is even, $n^2 + n + 6$ is even for, n even,

$$\begin{aligned} (2k)^2 &= 2^2 \cdot k^2 \\ &= 4k^2 \end{aligned}$$

$\rightarrow$ we take it as q

Case: 2,

suppose, $n \equiv \text{odd}$,

from Defⁿ, $n = 2k+1$ [$k \equiv \text{integer}$]

Now,

$$\boxed{n^2 + n + 6} = (2k+1)^2 + 2k+1 + 6$$

$$= 4k^2 + 4k + 1 + 2k + 1 + 6$$

$$= \boxed{4k^2 + 6k + 8} \rightarrow \text{int}$$

$$= 2 \boxed{(2k^2 + 3k + 4)} \rightarrow \text{int}$$

we take

$$2(2k^2 + 3k + 4) = q$$

$$\therefore p = 2q$$

$\therefore p$ is even, $n^2 + n + 6$ is even for, $n \equiv \text{odd}$

Proposition:

Every multiple of 4 equals to $1 + (-1)^n (2n-1)$

for $\boxed{\text{some } n \in \mathbb{N}} \rightarrow p$

$\rightarrow q$

Proposition: two integers have opposite parity if their sum is odd,

Proof:

- P : two integers have opposite parity
- Q : sum is odd,

Proof: suppose, m and n two integers with opposite parity.

Case 1: $m = \text{even}$, $n = \text{odd}$

from Defⁿ: $m = 2k_1$, $n = 2k_2 + 1$ $\{k_1, k_2 : \text{integers}\}$

$$\begin{aligned} m+n &= 2k_1 + (2k_2 + 1) = (2k_1 + 2k_2) + 1 \\ &= 2(k_1 + k_2) + 1 \quad \xrightarrow{\text{integer}} \\ &= 2q + 1 \end{aligned}$$

$q = k_1 + k_2$ (an integer)

$$\therefore m+n = 2q+1$$

from Defⁿ: $m+n$ is odd,

For $m: \text{even}$, $n: \text{odd}$, $m+n$ is odd

Case 2: $m: \text{odd}$, $n: \text{even}$

from, $m = 2k_1 + 1$, $n = 2k_2$ $\{k_1, k_2 : \text{integers}\}$

$$m+n = 2k_1 + 1 + 2k_2 = 2k_1 + 2k_2 + 1$$

$$= 2(k_1 + k_2) + 1 \quad \xrightarrow{\text{integer}}$$

$\therefore q = k_1 + k_2$ [integer]

$m+n = 2q+1$, , $m+n$ is odd
for, $m: \text{odd}$, $n: \text{even}$

Case: 1 (without the loss of generality)

$$m = \text{even}, \quad n = \text{odd}$$

some theorems with gcd:

Theorem: Bezout's identity:

if a and b are positive integers, there exists integers k and l

$$\gcd(a, b) = a \cdot k + l \cdot b$$

illustration:

$$a = 12, \quad b = 20$$

$$\gcd(12, 20) = 4$$

$$4 = 12 \cdot k + 20 \cdot l$$

- $k = 2, \quad l = -1,$

- $k = -3, \quad l = 2$

Sketch of proof:

- start with a and b positive integers (fixed)

- $\gcd(a, b)$ is also fixed

- we can vary k and l

$k \in \mathbb{Z} \text{ (free, integers)}$

$k \cdot a + l \cdot b \text{ : (integers)}$

$$a, b \in \mathbb{Z}$$

$$2a + 3b$$

We choose k and l in such a way such that $k \cdot a + l \cdot b$ is the smallest possible positive integer,

$$d = k \cdot a + l \cdot b \quad [d: \text{smallest possible integer}]$$

$d = \gcd(a, b)$ — we have to show

(i) d is a common divisor

(ii) d is the greatest common divisor

Proof: Assume a and b positive integer

for x and y integers,

$ax + by$ is integer

We can choose $x = k$ and $y = l$ in such a way such that,

$a \cdot l + b \cdot k$ gives the smallest possible positive integer,

$\therefore d = a \cdot k + b \cdot l$ [d : that number]

We need to prove,

$$d = \gcd(a, b)$$

(i) d is a common divisor:

first we have to show,

d|a:

by division algorithm,

$$a = dq + r \quad / \quad 0 \leq r < d$$

$$\rightarrow r = a - dq$$

$$r = a - (ak + lb)q$$

$$r = a(1 - kq) + b(-lq)$$

r is also,

$$r = ax + by$$

we said, d: smallest possible positive integer
constructed via $ax + by$

we also know,

$$0 \leq r < d$$

$\therefore r = 0$ from division algorithm

$$a = dq + 0 \rightarrow a = dq$$

$$\therefore d|a$$

similarly, $d|b$

$\therefore d$ is common divisor

Part: 2 Assume d' is some other common divisor of
a and b

$$\therefore d'|a \quad \text{and} \quad d'|b$$

$$\therefore a = d'm$$

$$b = d'n \quad [m, n \text{ integer}]$$

we know,

$$d = a \cdot k + b \cdot l$$

$$= d'm \cdot k + d'n \cdot l$$

$$= d' (m \cdot k + n \cdot l)$$

$$d' = \frac{d}{m \cdot k + n \cdot l}$$

$$\bullet m \cdot k + n \cdot l \leq -1 \quad , \bullet m \cdot k + n \cdot l \geq 1$$

$$\bullet m \cdot k + n \cdot l = 0 \quad [\text{discarded by the Def'n of GCD}]$$

$$d' : \gcd(a, b)$$

$$d = ax + by \quad (x, y : \text{integers})$$

$$\therefore \gcd(a, b) = ax + by \quad [x, y : \text{integers}]$$